MILITARY SPECIFICATION

RANGE FINDER, LASER: AN/VVS-1

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE EXERCISED IN OPERATING AND TESTING THE RANGE FINDER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL, PAINT, ETC. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE THE RANGE FINDER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 This specification covers one type of range finder designated as Laser Range Finder AN/VVS-1 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-R-14865	Receiver-Transmitter, Laser: 10559281
MIL-C-14866	Control, Power Supply: 10559261

FSC 1240

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MIL-C-14867	Control, Gunners, Laser Ranging: 10559221
MIL-I-14868	Indicator, Digital, Display: 10559241
MIL-I-14869	Indicator, Command, Display: 10559201

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10559401	Range Finder, Laser AN/VVS-1

PACKAGING DATA SHEET

10559401	Packaging of Range Finder, Laser AN/VVS-1: 10559401

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The range finder shall be manufactured in accordance with Drawing F10559401 and drawings pertaining thereto. The following components and cables shall each have met the requirements of their detailed specifications and drawings, respectively, prior to assembly as a range finder.

(a) Receiver-Transmitter, Laser: 10559281 - MIL-R-14865(MU)
(b) Control, Power Supply: 10559261 - MIL-C-14866(MU)
(c) Control, Gunners, Laser Ranging: 10559221 - MIL-C-14867(MU)
(d) Indicator, Digital Display: 10559241 - MIL-I-14868(MU)

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(e) Indicator, Command Display: 10559201 - MIL-I-14869(MU)
(f) Cable, Interconnect - 6W1: 10559286
(g) Cable, Interconnect - 6W3: 10559288
(h) Cable, Interconnect - 6W4: 10559289
(i) Cable, Interconnect - 6W5: 10559290

3.2 General specification.- The range finder shall meet the following requirements, where applicable, of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes

3.3 Performance.-

3.3.1 Range finder operation +60°F to +90°F.- The range finder shall meet the requirements of 3.3.1.1 through 3.3.1.15 when thermally stabilized within the temperature range of +60°F to +90°F and when the loads, power, and signals of Table I are applied as specified.

TABLE I

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
1.	Loads:		Connected to the following pins of 6W5 cable with return to P2-T:
1.1	Resistor	300 ohms ± 5%	P2-A
1.2	Resistor	300 ohms ± 5%	P2-B
1.3	Resistor	300 ohms ± 5%	P2-C
1.4	Resistor	300 ohms ± 5%	P2-D
1.5	Resistor	300 ohms ± 5%	P2-E
1.6	Resistor	300 ohms ± 5%	P2-F
1.7	Resistor	300 ohms ± 5%	P2-G
1.8	Resistor	300 ohms ± 5%	P2-H
1.9	Resistor	300 ohms ± 5%	P2-J
1.10	Resistor	300 ohms ± 5%	P2-K
1.11	Resistor	300 ohms ± 5%	P2-L
1.12	Resistor	240 ohms ± 5%	P2-P
1.13	Resistor	620 ohms ± 5%	P2-R

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TABLE I (continued)

ITEM	CONDITION	CHARACTERISTICS	CONNECTIONS
2.	Power sources:		Connected between following terminals:
2.1	24 Vdc	Tolerance: ± 2 Vdc	6W4P3(+) and 6W4P4(-)
3.	Signal sources:		
3.1	Laser On	24 ±2 Vdc	Applied to 6W5P2-M(+) with return to 6W5P2-T(-) unless otherwise specified.
3.2	Gunners Control	24 ±2 Vdc	Applied to 6W5P2-N(+) with return to 6W5P2-T(-) when specified herein.
4.	Auxiliary Equipment:		
4.1	Cable		Connected as specified on Figure 1.

3.3.1.1 Turn-on condition.- When the "Laser On" signal is first applied to the range finder, the following conditions shall exist:

(a) The range and range return nixie indicators shall read zero and the LAST button shall be illuminated.

(b) The RESET button and the upper (white) portion of the RANGE button on the controlling unit (the Gunners Laser Ranging Control or the Command Display Indicator) shall be illuminated.

(c) The mode indicator light shall illuminate for each selected switch position of the Laser Mode Control on the controlling unit.

(d) The FEED button on each control unit shall be continuously illuminated.

(e) The lower (red) portion of the RANGE button on the controlling unit shall begin to flash within 6 seconds.

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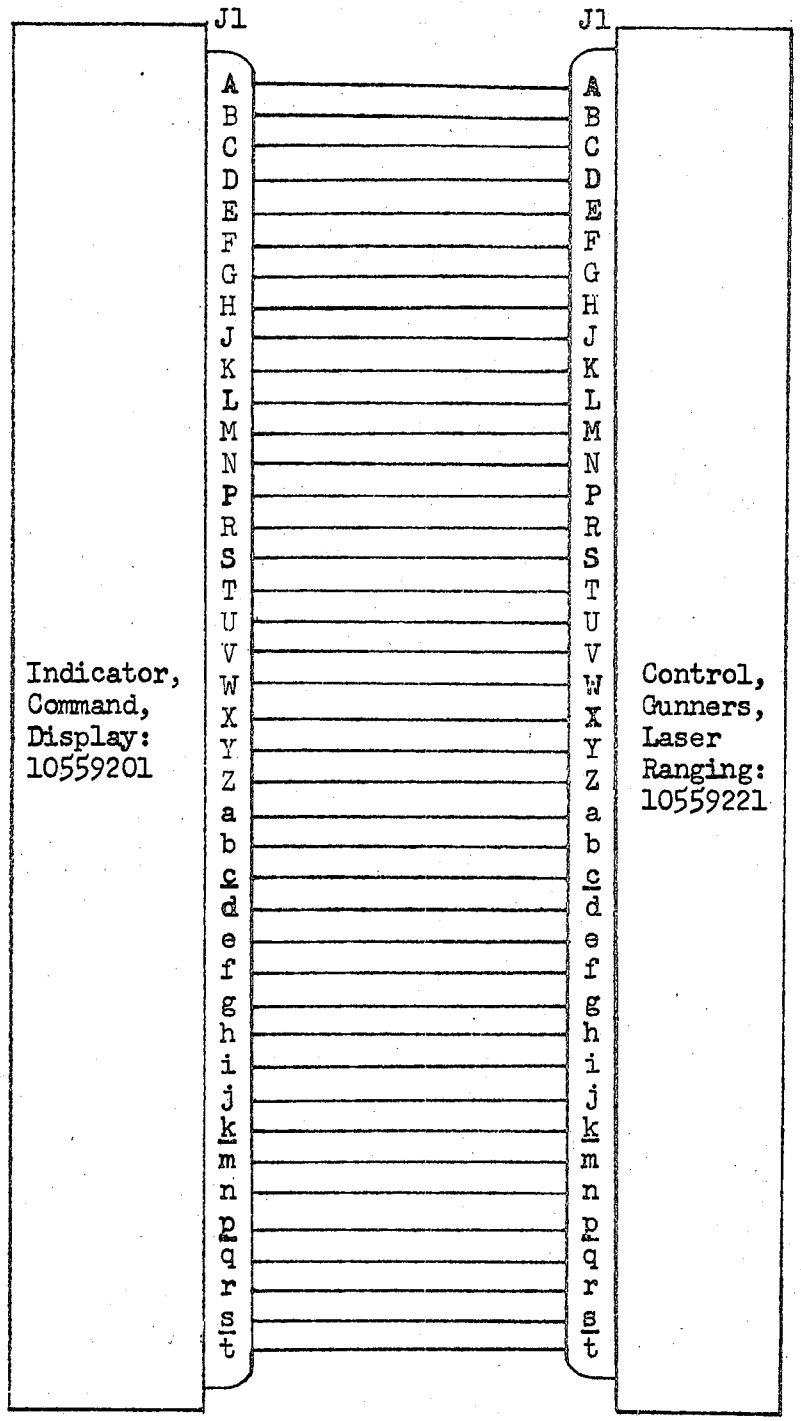


Figure 1 - Auxiliary Cable Connections

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3.3.1.2 PFN charge and flashtube firing.- Pressing the RANGE button when it is flashing (ranging) shall cause the PFN to discharge through the flashtube The flashing red light on the RANGE button shall turn off during ranging and turn on steady after ranging, indicating that the pulse forming network (PFN) is recharged. Pressing the RANGE button after ranging when the red light is steady shall not affect the flashtube. When the PFN is charged, pressing the RESET button shall cause the red light on the RANGE button to flash and the range return selector LAST indicator lamp shall light.

3.3.1.3 Target selection.-

(a) Pressing the "1" or "2" range return selector button shall cause the range of the first or second target, respectively, to be displayed on the nixie tubes. When there are one, two, or three target returns, pressing the LAST range return selector button shall cause the range to the last target to be displayed. When there are four or more target returns, pressing the LAST range return selector button shall cause the range to the third target to be displayed.

(b) When the FEED button is pressed, the GO (green) lamp shall light.

(c) When the Laser Mode Control switch is in the TEST position, the GO lamp shall light except during the following conditions when the SELECT lamp shall light:

(1) When the range return selector "1" button is depressed and more than two target echoes are received.

(2) When the range return selector "2" button is depressed.

(3) When the range return selector LAST button is depressed and four or more target echoes are received.

(4) When the target range is equal to or greater than the maximum range.

(d) When the green indicator light is on, the range return selector shall be inactivated and it shall be impossible to interrupt this connection except by pressing the RESET button.

3.3.1.4 RESET switch.- Whenever the RESET button is pressed, the range return nixie indicator shall read zero, the four range nixie indicators shall read zero, and the green or yellow indicator light shall also turn off.

3.3.1.5 LASER MODE CONTROL switch.-

3.3.1.5.1 TEST position.- Pressing the RANGE button when it is flashing shall not discharge the PFN through the flashtube. In addition, the requirements of 3.3.1.3 shall apply.

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3.3.1.5.2 ON position.- When the laser mode control switch is in the ON position, after ranging, the SELECT lamp shall light. Pressing the FEED button shall cause the GO lamp to light.

3.3.1.5.3 AUTO position.- When the laser mode control switch is in the AUTO position, after ranging, the GO lamp shall light. When no echoes are received the SELECT lamp shall light.

3.3.1.6 Ranging - laser firing.-

3.3.1.6.1 Minimum range, maximum range, ranging accuracy, and range resolution.- Minimum range, maximum range, ranging accuracy, and range resolution shall be as specified in MIL-R-14865.

3.3.1.6 2 Minimum time between ranging.- The requirements for minimum time between ranging shall be as specified in MIL-R-14865.

3.3.1.6.3 Maximum sustained ranging.- The maximum number of rangings shall be as specified in MIL-R-14865.

3.3.1.6.4 Remote ranging.- The requirements for remote ranging shall be as specified in MIL-R-14865.

3.3.1.7 Events counter.- The events counter shall perform as specified in MIL-R-14865.

3.3.1.8 Input voltage variation.- The range finder shall meet the ranging accuracy specified in 3.3.1.6.1 when the input voltage of the prime power source varies between +18 Vdc and +30 Vdc.

3.3.1.9 Maximum power drain.- The direct current drain from the prime power source shall be less than 12 amperes at any time.

3.3.1.10 Primary power transient.- The range indicator readout shall not change under the following conditions:

(a) When the input voltage is dropped to a value of 6 volts or less for a duration of at least 100 milliseconds.
(b) When the input voltage is increased to 90 volts for a duration of 100 milliseconds.

3.3.1.11 Peak output power.- The peak optical power output shall be as specified in MIL-R-14865.

3.3.1.12 Malfunction indication.-

3.3.1.12.1 Overvoltage.- If an overvoltage condition occurs in the PFN circuitry, the numbers "0" and "6" shall appear on the fourth range counter nixie indicator and the MALF lamp shall light.

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3.3.1.12.2 "A" trigger and counter.- If a malfunction occurs in the reply gating or counter circuit card, the numbers "0", "7", and "8", respectively, shall appear on the fourth range counter nixie indicator and the MALF lamp shall light.

3.3.1.12.3 Command signal line (to computer).- If a malfunction occurs in the command signal line to the computer, the numbers "0" and "9" shall appear on the fourth range counter nixie indicator and the MALF lamp shall light.

3.3.1.13 Data line (to computer).- All data line signals to the computer shall be as specified in MIL-R-14865.

3.3.1.14 Command signal output (to computer).- The boresight and command signals fed to the computer shall be as specified in MIL-R-14865.

3.3.2 Environmental.-

3.3.2.1 Storage temperature.- The range finder shall meet the requirements of 3.3.1.3 [except subparagraph (c)(4)] , 3.3.1.6.1 (maximum range only), and 3.3.1.11 at ambient temperature (60°F to 90°F) after exposure and thermal stabilization at -80°F and +160°F.

3.3.2.2 Operating temperature.- The range finder shall meet the requirements of 3.3.1.3 [except subparagraph (c)(4)], 3.3.1.6.1, 3 3.1.8, 3.3.1.10, and 3.3.1.11 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which it shall meet the requirements of 3.3.1.3 [except subparagraph (c)(4)], 3.3.1.6.1 (maximum range only), and 3.3.1.11 at ambient temperature (60°F to 90°F).

3.3.2.3 Vibration.- The range finder shall meet the requirements of 3.3.1.3 [except subparagraph (c)(4)] and shall have no physical failure during the following vibratory exposure:

(a)	Amplitude:	.5 inch maximum
(b)	Acceleration:	4 g maximum
(c)	Crossover frequency:	12.5 hertz (Hz)
(d)	Sweep time:	5 to 500 to 5 Hz in 15 minutes. Two sweeps in each of three mutually perpendicular axes.

Subsequent to the vibration exposure, the range finder shall meet the requirements of 3.3.1.6.1 (maximum range only) and 3.3.1.11.

3.3.3 Final performance.- After exposure to the environmental conditions of 3.3.2, the range finder shall meet the requirements of 3.3.1.1 through 3.3.1.14, except 3.3.1.3 (c)(4) and 3.3.1.13.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) range finders. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.-

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.-

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Functional tests.- The requirements and tests of Table II shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F to 90°F).

TABLE II

USE 100 PERCENT INSPECTION

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
101	Turn-on condition	3.3.1.1	4.6.1.1
102	PFN charge and laser firing	3.3.1.2	4.6.1.2
103	Target selection	3.3.1.3	4.6.1.3
104	RESET switch	3.3.1.4	4.6.1.4
105	LASER MODE CONTROL switch	3.3.1.5	4.6.1.5
106	Ranging -- laser firing	3.3.1.6	4.6.1.6
107	Events counter	3.3.1.7	4.6.1.7
108	Input voltage variation	3.3.1.8	4.6.1.8
109	Maximum power drain	3.3.1.9	4.6.1.9
110	Primary power transient	3.3.1.10	4.6.1.10
111	Peak output power	3.3.1.11	4.6.1.11
112	Malfunction indication	3.3.1.12	4.6.1.12
113	Data line (to computer)	3.3.1.13	4.6.1.13
114	Command signal output (to computer)	3.3.1.14	4.6.1.14

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III and IV herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot. Data obtained from lower level inspection tests may be applied toward fulfilling the requirements of the following characteristics of Table II.

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(a) Target selection [3.3.1.3(c)(4) only]
(b) Ranging - laser firing
(c) Events counter
(d) Peak power output
(e) Data line (to computer)
(f) Command signal output (to computer)

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.-

4.4.1 General.- One range finder shall be selected at random by a Government representative as a control sample from each 100 produced or from each months production, whichever occurs first. The control sample shall meet the requirements and tests in Table III.

TABLE III

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
301 Fabrication	3.1	Applicable drawings -- visual
302 General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental.- Three range finders shall be selected at random by a Government representative as control samples from each 50 produced or from each months production, whichever occurs first. The control samples shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table IV.

TABLE IV

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
303 Storage temperature	3.3.2.1	4.6.2.1
304 Operating temperature	3.3.2.2	4.6.2.1
305 Vibration	3.3.2.3	4.6.2.2

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4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.-

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2, the following test equipment shall be utilized in performance of the applicable test specified in 4.6.

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NOMENCLATURE	DESCRIPTION
1. Oscilloscope with camera	Frequency response dc to 150 megahertz; input impedance 1 megohm or greater; sensitivity 5 mv/cm; sweep time .05 microseconds per centimeter; camera lens f/1.3 (Tektronix C40 or equivalent); film ASA equivalent of 10,000 (Polaroid Type 410 or equivalent).
2. DC power supply	0 to 90 volts dc, 15 amps, Ripple: .1 volts rms max
3. Voltmeter DC	Range: 0 to 100 volts, Accuracy: ±.3% Input impedance: ⩾ 1 megohm
4. Ammeter DC	Range: 0 to 15 amps Accuracy: ±5%
5. Thermopile	TRG 108, Tekmar Instruments, Inc. or equivalent
6. Energy meter	TRG 113, Tekmar Instruments, Inc. or equivalent
7. Stop watch	.5 second dial divisions; 60 seconds per revolution of long hand; ±10% accuracy
8. Test set	Provides decimal readout on nixie tubes from 8-4-2-1 binary coded decimal inputs.

4.6 Test methods and procedures.- The following limitations are recommended for the protection of personnel against potential hazards from backscattered laser radiation.

(a) No tests shall be conducted in rain, snow or hail.

(b) Testing at night or day twilight should be avoided; however, if necessary, all personnel using the range finder, or in the vicinity, shall be equipped with a laser safety glass of at least density 4 at 6943 angstroms.

(c) Testing in daytime in heavy fog (visibility less than 200 meters) should be avoided; otherwise, safety glasses as specified above should be employed.

4.6.1 Range finder operation.- Connect the loads and power indicated in Table I and unless otherwise specified, allow the equipment to stabilize at a temperature between +60°F and +90°F.

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4.6.1.1 Turn-on condition.- Apply the "Laser On" signal of Table I as specified and observe the lighting of the push-button switches on each control, and the nixie indicator readout and the lighting of each position of the Laser Mode Control on the Command Display Indicator. Remove the "Laser On" signal, then apply the "Gunners Control" signal and reapply the "Laser On" signal as specified in Table I. Repeat the above observations. The panel display lighting and nixie indications for each turn-on condition shall be as specified in 3.3.1.1.

4.6.1.2 PFN charge and laser firings.- When the red RANGE light is flashing (system armed), press the RANGE button. The flashtube shall flash as specified in 3.3.1.2 which is evidenced by an audible sound caused by discharging of the PFN through the flashtube. Press the RANGE button again. The flashtube shall not flash (absence of sound from the flashtube). The RANGE button lighting sequence shall be as specified in 3.3.1.2. Press the RESET button. The lighting of the RANGE button and LAST indicator lamp shall be as specified in 3.3.1.2.

4.6.1.3 Target selection.- Set the emergency power switch (S2) to the OFF position. Perform steps 1 through 21 of Table V. The displays on the Command Display Indicator and Digital Display Indicator shall be as specified in Table V, thereby meeting the requirements of 3.3.1.3, except 3.3.1.3(c)(4). The test method and procedure for 3.3.1.3(c)(4) shall be as specified in MIL-R-14865.

4.6.1.4 RESET switch.- Performance of step 9, 15, or 21 of Table V (see 4.6.1.3) shall constitute meeting the requirements of 3.3.1.4.

4.6.1.5 LASER MODE CONTROL switch.-

4.6.1.5.1 TEST position.- Perform steps 1 and 2 of Table V. There shall be no audible sound from the flash tube, thereby meeting the requirements of 3.3.1.5.1.

4.6.1.5.2 ON position.- Perform steps 1 through 4 of Table VI. The displays on the Command Display Indicator and Digital Display Indicator shall be as specified in Table VI, thereby meeting the requirements of 3.3.1.5.2.

4.6.1.5.3 AUTO position.- Perform steps 5, 6, and 7 of Table VI. The displays on the Command Display Indicator and Digital Display Indicator shall be as specified in Table VI, thereby meeting the requirements of 3.3.1.5.3.

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TABLE V

Inputs (Unless Otherwise specified, use Command Display Indicator)		Indications					
Step Number	Press and Release Switches as Indicated	RANGE RETURN SELECTOR Lights	Range Returns	Range* (Nixies) (±5 Meters)	GO Light (Green)	SELECT Light (Yellow)	RANGE RETURN SELECTOR Inactivated
Set LASER MODE CONTROL to TEST position							
1		LAST	0	0000	Off	Off	No
2	RANGE	LAST	1	0830	On	Off	Yes**
3	RANGE	LAST	2	1835	On	Off	Yes
4	RANGE	LAST	3	2830	On	Off	Yes
5	RANGE	LAST	4	2830	Off	On	No**
6	"1"	"1"	4	0830	Off	On	No
7	"2"	"2"	4	1835	Off	On	No
8	FEED	"2"	4	1835	On	Off	Yes
9	RESET	LAST	0	0000	Off	Off	No
End of first test sequence							
10	"1"	"1"	0	0000	Off	Off	No
11	RANGE	"1"	1	0830	On	Off	Yes
12	RANGE	"1"	2	0830	On	Off	Yes

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TABLE V (Continued)

Inputs (Unless Otherwise specified, use Command Display Indicator)		Indications					
Step Number	Press and Release Switches as Indicated	RANGE RETURN SELECTOR Lights	Range Returns	Range* (Nixies) (±5 Meters)	GO Light (Green)	SELECT Light (Yellow)	RANGE RETURN SELECTOR Inactivated
13	RANGE	"1"	3	0830	Off	On	No
14	FEED	"1"	3	0830	On	Off	Yes
15	RESET	LAST	0	0000	Off	Off	No
	End of second test sequence						
16	"2"	"2"	0	0000	Off	Off	No
17	RANGE	"2"	1	9995***	Off	On	No
18	FEED	"2"	1	9995	Off	On	No
19	RANGE	"2"	2	1835	Off	On	No
20	FEED	"2"	2	1835	On	Off	Yes
21	RESET	LAST	0	0000	Off	Off	No
	End of third test sequence						

* These range indications shall also appear on the test set when connected to data line to computer.

** Verify by pressing "1", "2" and LAST buttons on RANGE RETURN SELECTOR. (Other items in this column need not be verified.)

*** If nixies read 0000, reset system and repeat test sequence. Test set indication shall be 1995.

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TABLE VI

Inputs (Unless otherwise specified, use Command Display Indicator)		Indications					
Step Number	Press and Release Switches as Indicated	RANGE RETURN SELECTOR Lights	Range Returns	Range (Nixies)	GO Light (Green)	SELECT Light (Yellow)	RANGE RETURN SELECTOR Inactivated
Select a target beyond minimum range and within maximum range which is large enough to intercept the total laser beam with nothing visible between LRF and target.							
Set LASER MODE CONTROL switch to ON position.							
1		LAST	0	0000	Off	Off	No
2	RANGE	LAST	1	Range to Target	Off	On	No
3	FEED	LAST	1	Range to Target	On	Off	Yes*
4	RESET	LAST	0	0000	Off	Off	No
Set LASER MODE CONTROL switch to AUTO position.							
5	RANGE	LAST	1	Range to Target No.3 ****	On	Off	Yes*
6	RESET	LAST	0	0000	Off	Off	No
7***	RANGE	LAST	0	9995**	Off	On	No
Change to Gunners Control (Apply the Gunner's Control signal) and repeat steps 1 through 7 using the Gunners Control.							

*Verify by pressing "1", "2", and LAST buttons on RANGE RETURN SELECTOR (Other items in this column need not be verified.).
**If nixies read 0000, reset system and repeat test sequence.
***Receiver aperture shall be covered.
****See MIL-R-14865.

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4.6.1.6 Ranging -- laser firing.-

4.6.1.6.1 Minimum range, maximum range, ranging accuracy, and range resolution.- The test method and procedure shall be as specified in MIL-R-14865.

4.6.1.6.2 Minimum time between ranging.- The test method and procedure shall be as specified in MIL-R-14865.

4.6.1.6.3 Maximum sustained ranging.- The test method and procedure shall be as specified in MIL-R-14865.

4.6.1.6.4 Remote ranging.- The test method and procedure shall be as specified in MIL-R-14865.

4.6.1.7 Events counter.- The test method and procedure shall be as specified in MIL-R-14865.

4.6.1.8 Input voltage variation.- Set the LASER MODE CONTROL switch to the ON position and perform five ranging operations against Target No. 3 (see MIL-R-14865) at each of the following prime voltages: 24 ±2 Vdc, 18 ±.5 Vdc, and 30 ±.5 Vdc. The ranging accuracy at each input voltage shall be as specified in 3.3.1.8.

4.6.1.9 Maximum power drain.- Measure the prime input current to the range finder while performing steps 1 and 2 of Table VI, using the ammeter specified in 4.5.3. The input current shall be within the limits specified in 3.3.1.9.

4.6.1.10 Primary power transient.- Perform steps 1 and 2 of Table V to obtain a range readout on the range indicator. Drop the primary input voltage to a value of 6 volts or less for a duration of at least 100 milliseconds and observe the range indicator readout. (Note: Do not interrupt the input voltage, item 2.1, Table I, for a period greater than 3 seconds.) Apply a primary input voltage transient of 90 volts for a duration of 100 milliseconds. Perform step 5 of Table VI using the Command Display Indicator, then reapply the 90-volt, 100-millisecond transient and observe the range indicator readout. The effect of the primary input voltage transients shall be as specified in 3.3.1.10.

4.6.1.11 Peak output power.- The test method and procedure shall be as specified in MIL-R-14865.

4.6.1.12 Malfunction indication.- Set the LASER MODE CONTROL switch to the TEST position and depress RESET button.

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4.6.1.12.1 Overvoltage.- Connect a 66.5K ±1% resistor between pins A and J on the test access connector J3. The malfunction indications shall be as specified in 3.3.1.12.1.

4.6.1.12.2 "A" trigger and counter.- Momentarily ground pin B on the test access connector J3. The malfunction indications shall be as specified in 3.3.1.12.2.

4.6.1.12.3 Command signal line (to computer).- Connect a 51-ohm ±5% resistor between pin P of the computer connector P2 and ground. The malfunction indications shall be as specified in 3.3.1.12.3.

4.6.1.13 Data line (to computer).- The test method and procedure shall be as specified in MIL-R-14865.

4.6.1.14 Command signal output (to computer).- The test method and procedure shall be as specified in MIL-R-14865.

4.6.2 Environmental tests.- The test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 except that the tolerance requirement for vibration frequency measuring instruments shall not apply.

4.6.2.1 Temperature test.-

(a) Place the range finder in the test chamber and reduce the chamber temperature to -80°F.
(b) Soak the range finder at this temperature for at least 5 hours with the equipment not operating.
(c) Raise the chamber temperature to -40°F.
(d) Following thermal stabilization at -40°F, place the range finder in operation and perform the tests specified in 4.6.1.3 [requirement 3.3.1.3(c)(4) excepted], 4.6.1.6.1, 4.6.1.8, 4.6.1.10, and 4.6.1.11.
(e) Raise the chamber temperature to +160°F.
(f) Soak the range finder at +160°F for at least 5 hours with the equipment not operating.
(g) Reduce the chamber temperature to +125°F.
(h) Following thermal stabilization at +125°F, place the range finder in operation and perform the tests specified in 4.6.1.3 [requirement 3.3.1.3(c)(4) excepted], 4.6.1.6.1, 4.6.1.8, 4.6.1.10, and 4.6.1.11.
(j) Reduce the chamber temperature to ambient.
(k) Following stabilization at ambient temperature, place the range finder in operation and perform the tests specified in 4.6.1.3 [requirement 3.3.1.3(c)(4) excepted], 4.6.1.6.1 (maximum range only), and 4.6.1.11.

The range finder shall meet the requirements of 3.3.2.1 and 3.3.2.2.

Note: The rate of temperature change shall not be greater than 18°F per minute.

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4.6.2.2 Vibration.- Mount the range finder firmly on the vibration table and perform the following:

(a) Test in accordance with Method 514-I of MIL-STD-810 for category (f) equipment, except that the acceleration shall not exceed ±4 g and the maximum excursion (double amplitude) shall be .5 inch.

(b) Sweep the frequency from 5 Hz to 500 Hz and back to 5 Hz in a period of 15 minutes. Perform two sweeps in each of the three mutually perpendicular axes.

(c) During each axis of vibration perform the tests specified in 4.6.1.3 [requirement 3.3.1.3(c)(4) excepted].

(d) Subsequent to vibration, examine the equipment for structural failures, loose hardware, wire breakage, and abrasion, and perform the tests specified in 4.6.1.6.1 (maximum range only) and 4.6.1.11.

The range finder shall meet the requirements of 3.3.2.3.

4.6.3 Final performance test.- After conclusion of the environmental tests of 4.6.2, perform the tests specified under 4.6.1, except 4.6.1.3 requirement 3.3.1.3(c)(4) and 4.6.1.13. The rangefinder shall meet the requirements of 3.3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10559401. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Laser Range Finder is intended for use in the M60A1E2 Tank Weapon Control System.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable packaging data sheet number (see 5.1).
(c) Selection of applicable levels of preservation, packaging, and packing.
(d) Applicable stock number.
(e) Requirement for First Article submission.

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